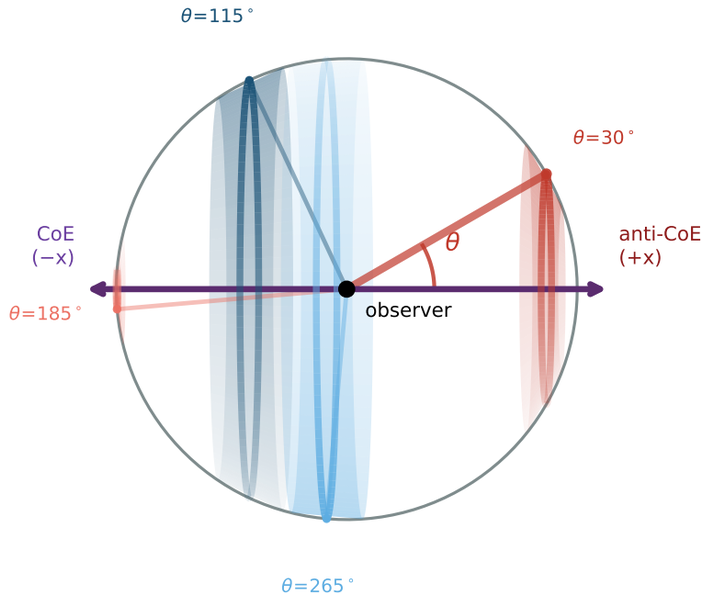
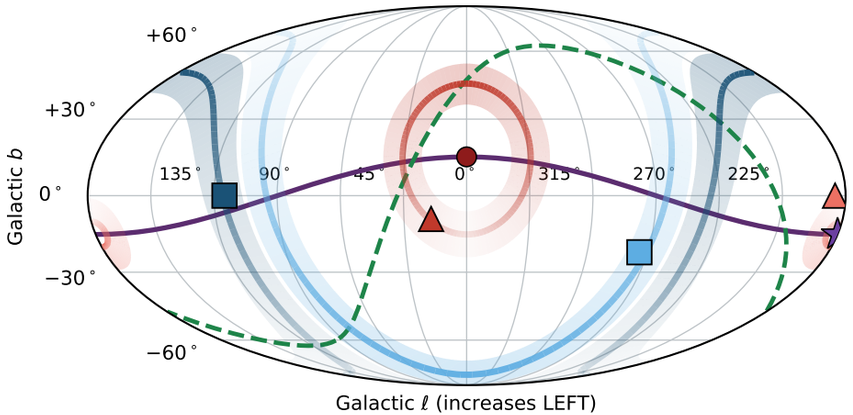
Pure CoE axisymmetry: full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite ($\alpha \propto 1 - 0.88|\psi|/\pi$)
(each 18° end-to-end = $\pm 9^\circ$; MAX θ=30°/185°; MIN θ=115°/265°; anti-CoE (0°, $+15^\circ$), CoE axis tilted $+15^\circ$)

A. Model frame — full loops, opacity gradient axisymmetric about x = CoE ↔ anti-CoE darkest at θ ray, fading to 12% opacity opposite; each 18° end-to-end ($\pm 9^\circ$)

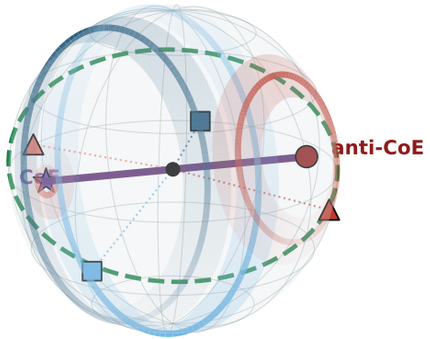


full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite
$\alpha \propto 1 - 0.88|\psi|/\pi$, ψ = ring azimuth from θ-ray point · each 18° end-to-end ($\pm 9^\circ$)
(after-2ℓ A(ℓ): two MAX + two MIN)

B. Migkas $A(\ell)$ loops on sky (anti-CoE at (0°, $+15^\circ$))
(Galactic Mollweide; CoE axis tilted $+15^\circ$; ℓ increases LEFT)
full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite
no roll about CoE (0°)



C. Sphere 3D — full loops
(opacity gradient darkest at θ ray, fading to 12% opacity opposite; celestial equator dashed)



MAX θ=30°, 185°; MIN θ=115°, 265° (full loops, gradient; 18° end-to-end = $\pm 9^\circ$)

CoE ↔ anti-CoE axis
MAX θ=30° full loop, gradient [21,39]°
MAX θ=185° full loop, gradient [176,194]°
MIN θ=115° full loop, gradient [106,124]°
MIN θ=265° full loop, gradient [256,274]°
celestial equator (Dec = 0)
MAX (17°, -9°)
MAX (185°, $+0^\circ$)
MIN (115°, $+0^\circ$)
MIN (274°, -22°)
anti-CoE (0°, $+15^\circ$)
CoE (180°, -15°)